INCH-POUND

MIL-DTL-45918A
18 May 2009
SUPERSEDING
MIL-C-45918
4 December 1968

DETAIL SPECIFICATION

CLEVISES AND CONNECTORS
(ROD END)

INACTIVE FOR NEW DESIGN AFTER 4 February 1999.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers adjustable and plain rod end clevises and rod end connectors.

1.2 Classification. Clevises and connectors are of the following types, styles, sizes, and compositions, as specified (see 6.2).

1.2.1 Types and styles.

Type I - Adjustable clevis
Style 1 - Wide fork (NASM27975 and NASM35812)
Style 2 - Narrow fork (NASM27975)

Type II - Plain clevis
Style 1 – Wide fork (NASM27976 and NASM35812)
Style 2 – Narrow fork (NASM27976)

Type III - Connector
Style 1 - Plain (MS35813)
Style 2 - Rigid linkage (A-A-52493)
Style 3 - Threaded (AS49011)

1.2.2 Sizes. Clevises and connectors are of the sizes listed in the applicable part standard.

1.2.3 Compositions. Clevises and connectors are made of the compositions listed on the applicable part standard and in accordance with 3.1.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia (DSCP), ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

AMSC N/A FSC 5340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:14Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 Screw Thread Standards for Federal Services

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-7742 Screw Threads, Standard, Optimum Selected Series: General Specification for (Inactive for new design)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 Identification Marking of U.S. Military Property

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

American Society of Mechanical Engineers (ASME)

ASME B46.1 Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available from www.asme.org or the American Society of Mechanical Engineers International, Three Park Avenue, M/S 10E New York, NY 10016-5990.)

American Society for Quality Control (ASQ)

ASQ Z1.4 Sampling Procedures and Tables for Inspection by Attributes

Copies of this document are available from www.asq.org American Society for Quality Control, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:14Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials Standard (ASTM)

ASTM A123/A123M	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM B633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM E8/E8M	Standard Test Methods for Tension Testing of Metallic Materials
ASTM E1282	Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials

(Copies of these documents are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

Society of Automotive Engineers (SAE)

SAE AIR4127	Steel: Chemical Composition and Hardenability
SAE AMS-S-6758	Steel, Chrome-Molybdenum (4130) Bars and Reforging Stock (Aircraft Quality)
SAE AMS-QQ-P-416	Plating, Cadmium (Electrodeposited)

(Copies of these documents are available from www.sae.org or the American Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Clevises and connectors shall be of the following compositions, as specified (see 6.2).

3.1.1 Composition A. Carbon steel shall be SAE 1010 or 1015-1029 or 1111 or equivalent in accordance with SAE AIR4127. These steels shall have a minimum ultimate tensile strength of 55,000 PSI (379.21 MPa).

3.1.2 Composition B. Carbon steel shall be SAE 1035 or 1042 in accordance with SAE AIR4127. These steels shall have a minimum ultimate tensile strength of 65,000 PSI (448.16 MPa).

3.1.3 Composition C. Alloy steel shall be SAE 4130 in accordance with SAE AMS-S-6758. This steel shall be heat treated to an ultimate tensile strength of 125,000-145,000 PSI (861.84 – 999.74 MPa).

3.2 Design and construction. The design and construction of clevises and connectors shall be as specified on the applicable part standard.

3.2.1 Dimensions and tolerances. Dimensions and tolerances shall be as specified on the applicable part standard.

3.3 Mechanical properties. Clevises and connectors shall be capable of withstanding the ultimate axial loads specified on the applicable part standard.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:14Z
Check the source to verify that this is the current version before use.

3.4 Protective finish. When specified (see 6.2), protective finishes shall be as follows.

3.4.1 Cadmium plating. Cadmium plating shall be Type II, Class 3 in accordance with SAE AMS-QQ-P-4l6.

3.4.2 Zinc coating. Zinc coating shall be Type II, Fe/Zn13 in accordance with ASTM B633 or hot dipped galvanized in accordance with ASTM A123/A123M.

3.5 Threads. The threads shall be as specified on the applicable part standard and shall be in accordance with FED-STD-H28 or MIL-S-7742, as applicable.

3.6 Surface roughness. Surface roughness shall be as specified on the applicable part standard and shall be in accordance with ASME B46.l.

3.7 Workmanship. Clevises and connectors shall be free of burrs, slivers, sharp edges, surface contamination, cracks and ruptures.

3.8 Marking. When specified (see 6.2), clevises and connectors shall be marked in accordance with MIL-STD-130.

4. VERIFICATION

4.1. Inspection provisions.

4.1 Lot. A lot shall consist of all clevises or connectors of the same type, style, size, composition and heat of steel, manufactured under essentially, the same conditions, and offered for acceptance at one time.

4.1.2 Sampling.

4.1.2.1 Sampling for examination. A random sample of clevises or connectors shall be taken from each lot in accordance with ASQ Z1.4, Inspection Level II.

4.1.2.2 Sampling for axial load test. A random sample of clevises or connectors shall be taken from each lot in accordance with ASQ Z1.4, Inspection Level S-4.

4.1.2.3 Sampling for tensile strength test. Two samples shall be taken from each lot for the tensile strength test.

4.1.2.4 Sampling for tests of protective finish. Sampling for tests of protective finish shall be in accordance with the applicable specification of 3.4.

4.1.2.5 Sampling for chemical analysis. One sample shall be taken from each lot for chemical analysis.

4.2 Examination. Each clevis or connector taken in accordance with 4.1.2.1 shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted as specified in Table I, as applicable. Any clevis or connector containing one or more defects shall be rejected, and if the number of defective clevises or connectors in the sample exceeds the acceptance number for that sample, the represented lot shall be rejected. Table I shall be applied separately to major and minor defects for purposes of lot acceptance or rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:14Z
Check the source to verify that this is the current version before use.

Table I

Classification of Defects

Categories	Defects	Inspection Method
Critical	None defined	
Major	AQL = 2.5	
101	Type, style, size and composition not as specified	Visual, SIE *
102	Threads not as specified	SIE
103	Jaw opening not as specified	SIE
104	Hole diameter not as specified	SIE
Minor	AQL = 6.5	
201	Length not as specified	SIE
202	Web thickness not as specified	SIE
203	Overall width of jaw not as specified	SIE
204	Body diameter not as specified	SIE
205	Boss diameter not as specified	SIE
206	Protective coating missing or incomplete	Visual
207	Surface roughness not as specified	SIE
208	Workmanship (3.7)	Visual
209	Marking of items not as specified	Visual

* Standard Inspection Equipment

4.3 Tests.

4.4.1 Axial load test. Each clevis and connector taken in accordance with 4.1.2.2 shall be subjected to an axial load test using the ultimate axial load values specified on the applicable part standard. After testing there shall be no evidence of stripped threads or other thread deformations (when applicable) and no signs of cracks or ruptures.

4.3.2 Tensile strength test. Samples selected in accordance with 4.1.2.3 shall be subjected to a tensile strength test in accordance with ASTM E8/E8M to determine conformance to the tensile strength requirements of 3.1.

4.3.3 Protective finish test. Samples selected in accordance with 4.1.2.4 shall be tested for protective finish in accordance with the applicable specification of 3.4. Acceptance/rejection criteria shall be in accordance with the applicable specification.

4.3.4 Chemical analysis. Samples selected in accordance with 4.1.2.5 shall be analyzed by chemical analysis per ASTM E1282. Mill certification that the material complies with the chemical composition requirements of 3.1 will be acceptable in lieu of the above test.

4.3.5 Any sample failing to pass the tests specified in 4.3.1, 4.3.2 or 4.3.4 shall be cause for rejection of the represented lot.

5. PACKAGING

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:14Z
Check the source to verify that this is the current version before use.

5.1 Packaging. For Acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contracting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Clevises and connectors are intended to be attached to a shaft or rod by welding, threading or pinning to provide a means of fastening, or attaching the shaft or rod to another part.

6.2 Ordering data. Procurement documents should specify the following.

(a) Title, number and date of this specification.
(b) Type, style and size required (see 1.2).
(c) Material (see 3.1).
(d) Applicable part standard (see 1.2).
(e) Protective coating, if required (see 3.4).
(f) Marking, if required (see 3.8).
(g) Packaging requirements (see 5.1).

6.3 Subject term (Key word) listing.

Cadmium
Narrow fork
Plain
Rigid linkage
Threaded wide fork

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AR
Navy - AS
Air Force - 99

Preparing activity;
DLA-IS

(Project No. 5340-2009-015)

Reviewer activities:
Army - AT, AV, CR4, GL, MI
Navy - MC, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change. You should verify the currency of the information above using the ASSIST online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:14Z
Check the source to verify that this is the current version before use.